100 Days of Code: Python Programming Study Notes

I. Introduction

* 100 days of code to learn Python programming from zero to job-ready
* Dedicated 100 days of learning with byte-sized lessons
* Progress from easy to complex projects used in industry

II. What is Python?

* **Definition**: Python is a dynamically typed, object-oriented programming language
* **Creator**: Created by Guido van Rossum in 1991
* **Name Origin**: Named after the TV show "Monty Python's Circus" that he was watching at the time

III. Key Features of Python

* **Easy to Learn**: Easy to learn and use
* **Fast Development**: Fast development and execution
* **Large Community**: Large community and extensive libraries

IV. Learning Objectives

* Learn Python programming from scratch
* Understand object-oriented programming concepts
* Work on complex projects used in industry

* Practice with exercises, quizzes, and challenges

V. Setting up Python

* **Installation**: Download and install Python from python.org
* **IDE/Text Editor**: Choose a preferred IDE or text editor (e.g., Replit, VS Code)
* **New Project**: Create a new project and start coding

VI. Replit: A Collaborative Browser-Based IDE

* **Online Coding**: Allows online coding and collaboration
* **Easy to Use**: Easy to use and share code with others
* **Suitable for All**: Suitable for those with slow computers or limited resources

VII. What is Programming?

* **Definition**: Programming is a way to tell computers what to do
* **Use**: Use programming to create instructions for computers to follow
* **Examples**: Examples: calculators, games, and software applications

VIII. Features of Python

* **Simple**: Python is easy to learn and understand
* **Platform-independent**: Python code can run on any platform (Windows, Linux, Mac)

* **Open-source**: Python is free to use and distribute
* **Large library support**: Python has a vast collection of libraries and modules
* **Easy to integrate with other languages**: Python can be linked to other programming languages

IX. Applications of Python

* **Data Science**: Python is used for data analysis, visualization, and machine learning
* **Web Development**: Python can be used to create web applications and web services
* **Database Management**: Python can be used to interact with databases
* **Machine Learning**: Python is used for building and training machine learning models
* **Complex Mathematical Operations**: Python can perform complex mathematical calculations

X. Installing Python

* **Installation**: Python can be installed on any platform (Windows, Linux, Mac)
* **Path Variable**: Pay attention to the path variable during installation to ensure Python can be used through the terminal

XI. Python Basics and Repl

* **Problem: Python Not Found Error**: Error occurs when typing "Python" in Linux or Mac
* **Solution**: Type "Python 3" to access Repl (Read-Evaluate-Print Loop)
* **Repl (Read-Evaluate-Print Loop)**: A tool to run Python code and see output
* **Working with Repl**: Write code in Repl, click "Run" to execute code, and see output in Repl

XII. Next Steps

* **Access Official Python Playlist**: Access the official Python playlist (100 days of code)
* **Like and Subscribe**: Like and subscribe to the channel to stay updated
* **Wait for Day 2**: Wait for day 2 to learn more about Python basics.